$$Info(D) = I(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940$$

$$Info_{age}(D) = 0.694$$

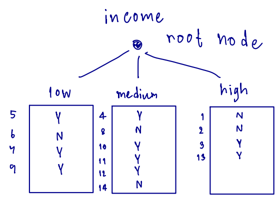
1. $Gain(age) = Info(D) - Info_{age}(D)$

$= 0.940 - 0.694 = 0.246$

Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31…40	high	no	fair	yes
4	>40	medium	no	fair	yes
5.	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31…40	medium	no	excellent	yes
13	31…40	high	yes	fair	yes
14	>40	medium	no	excellent	no

Note: The data set is adapted from



income	p_i	n_i	$I(p_i,n_i)$
low	3	1	0.8113
medium	4	2	0.9182
high	2	2	1

$$I(3,1) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$$

$$I(4,2) = -\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2\left(\frac{2}{6}\right)$$

$$I(2,2) = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$$Info_{income} = \frac{4}{14}(0.8113) + \frac{6}{14}(0.9182) + \frac{4}{14}(1) = 0.9110$$

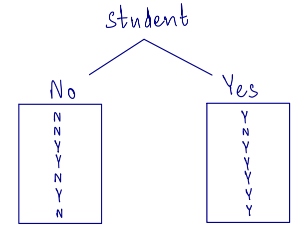
2. $Gain(income) = 0.940 - 0.9110$

$= 0.029$

Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from



Student	p_i	n_i	$I(p_i,n_i)$
No	3	4	0.9852
Yes	6	1	0.5917

$$I(3,4) = -\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{4}{7}\log_2\left(\frac{4}{7}\right)$$

$$I(6,1) = -\frac{6}{7}\log_2\left(\frac{6}{7}\right) - \frac{1}{7}\log_2\left(\frac{1}{7}\right)$$

$$Info_{student}(D) = \frac{7}{14}I(3,4) + \frac{7}{14}I(6,1)$$

$$= \frac{7}{14}(0.9852) + \frac{7}{14}(0.5917)$$

$$= 0.7885$$

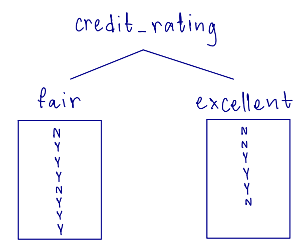
3. $Gain(student) = 0.940 - 0.7885$

$= 0.1515$

Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from



credit_rating	p_i	n_i	$I(p_i,n_i)$
fair	6	2	0.8113
excellent	3	3	1

$$I(6,2) = -\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right)$$

$$I(3,3) = -\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{3}{6}\log_2\left(\frac{3}{6}\right)$$

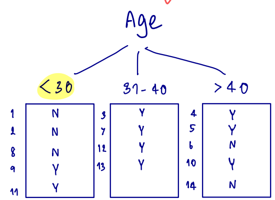
$$Info_{(credit\_rating)} = \frac{8}{14}I(6,2) + \frac{6}{14}I(3,3)$$

$$= \frac{8}{14}(0.8113) + \frac{6}{14}(1)$$

$$= 0.8922$$

4. $Gain(credit\_rating) = 0.940 - 0.8922$

$= 0.0478$



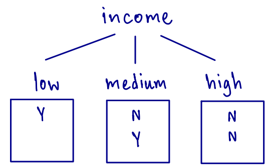
Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from

$$Info(D) = I(2,3) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$$

$$Info(D) = 0.9705$$



income	p_i	n_i	$I(p_i,n_i)$
low	1	0	0
medium	1	1	1
high	0	2	0

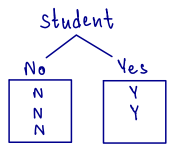
$$Info_{income}(D) = \frac{1}{5}I(1,0) + \frac{2}{5}I(1,1) + \frac{2}{5}I(0,2)$$

$$= \frac{1}{5}(0) + \frac{2}{5}(1) + \frac{2}{5}(0)$$

$$= 0.8$$

$$Gain(income) = 0.9701 - 0.8$$

$$= 0.1701$$



student	p_i	n_i	$I(p_i,n_i)$
No	3	0	0
Yes	2	0	0

$$Info_{student}(D) = \frac{2}{5}I(2,0) + \frac{3}{5}I(2,0)$$

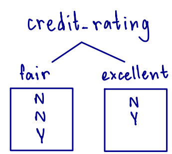
$$= \frac{2}{5}(0) + \frac{2}{5}(0)$$

$$= 0$$

$$Gain(student) = 0.9701 - 0$$

$$= 0.9701$$

เลือก Student



credit_rating	p_i	n_i	$I(p_i,n_i)$
fair	1	2	0.9183
excellent	1	1	1

$$Info_{credit\_rating} = \frac{3}{5}I(1,2) + \frac{2}{5}I(1,1)$$

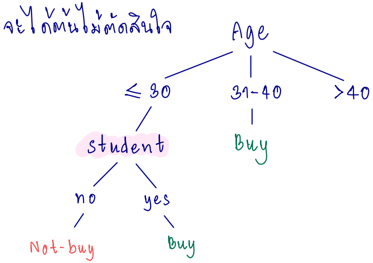
$$= \frac{3}{5}(0.9183) + \frac{2}{5}(1)$$

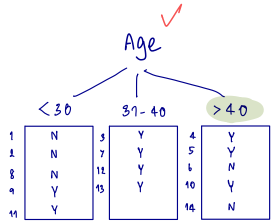
$$= 0.9510$$

$$Gain(credit\_rating) = 0.9701 - 0.9510$$

$$= 0.0191$$

จะได้ต้นไม้ตัดสินใจ





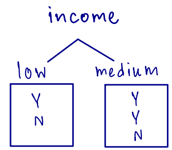
Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Note: The data set is adapted from

$$Info(D) = I(3,2) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

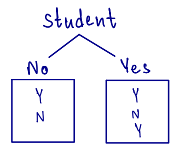
$$Info(D) = 0.9710$$



student	p_i	n_i	$I(p_i, n_i)$
low	1	1	1
medium	2	1	0.9183

$$Info_{income}(D) = \frac{2}{5} I(1,1) + \frac{3}{5} I(2,1)$$
$$= \frac{2}{5}(1) + \frac{3}{5}(0.9183)$$
$$= 0.9510$$

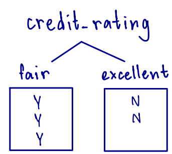
$$Gain(income) = 0.9710 - 0.9510$$
$$= 0.0200$$



student	p_i	n_i	$I(p_i, n_i)$
No	1	1	1
Yes	2	1	0.9183

$$Info_{student}(D) = \frac{2}{5} I(1,1) + \frac{3}{5} I(2,1)$$
$$= \frac{2}{5}(1) + \frac{3}{5}(0.9183)$$
$$= 0.9510$$

$$Gain(student) = 0.9710 - 0.9510$$
$$= 0.0200$$



credit_rating	p_i	n_i	$I(p_i, n_i)$
fair	3	0	0
excellent	0	2	0

$$Info_{credit\_rating} = \frac{3}{5} I(3,0) + \frac{2}{5} I(0,2)$$
$$= \frac{3}{5}(0) + \frac{2}{5}(0)$$
$$= 0$$

$$Gain(credit\_rating) = 0.9710 - 0$$
$$= 0.9710$$

เลือก credit_rating

จะได้ต้นไม้ตัดสินใจ